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**DIPARTIMENTO
DI INFORMATICA**

TLN:
WordNet

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credits

il materiale di questa lezione è tratto dagli articoli:

- G. A. Miller. Wordnet: a lexical database for English. *Commun. ACM*, 38(11):39–41, 1995.
- G. A. Miller, R. Beckwith, C. Fellbaum, D. Gross, and K. J. Miller. Introduction to WordNet: An On-line Lexical Database. *International Journal of Lexicography*, 4(3):235–244, 1993.
- G. A. Miller, "Nouns in WordNet: A Lexical Inheritance System", *International Journal of Lexicography*, 4(3):245–264, 1993.
- Christiane Fellbaum, "English Verbs as a Semantic Net", *International Journal of Lexicography*, 4(3):278–301, 1993.

an on-line lexical database

- WordNet is an **on-line lexical reference system** whose design is inspired by *psycholinguistic theories* on human lexical memory.
- English nouns, verbs, and adjectives are organized into **synonym sets**, each representing one underlying lexical *concept*.
 - different relations link the synonym sets.

problem

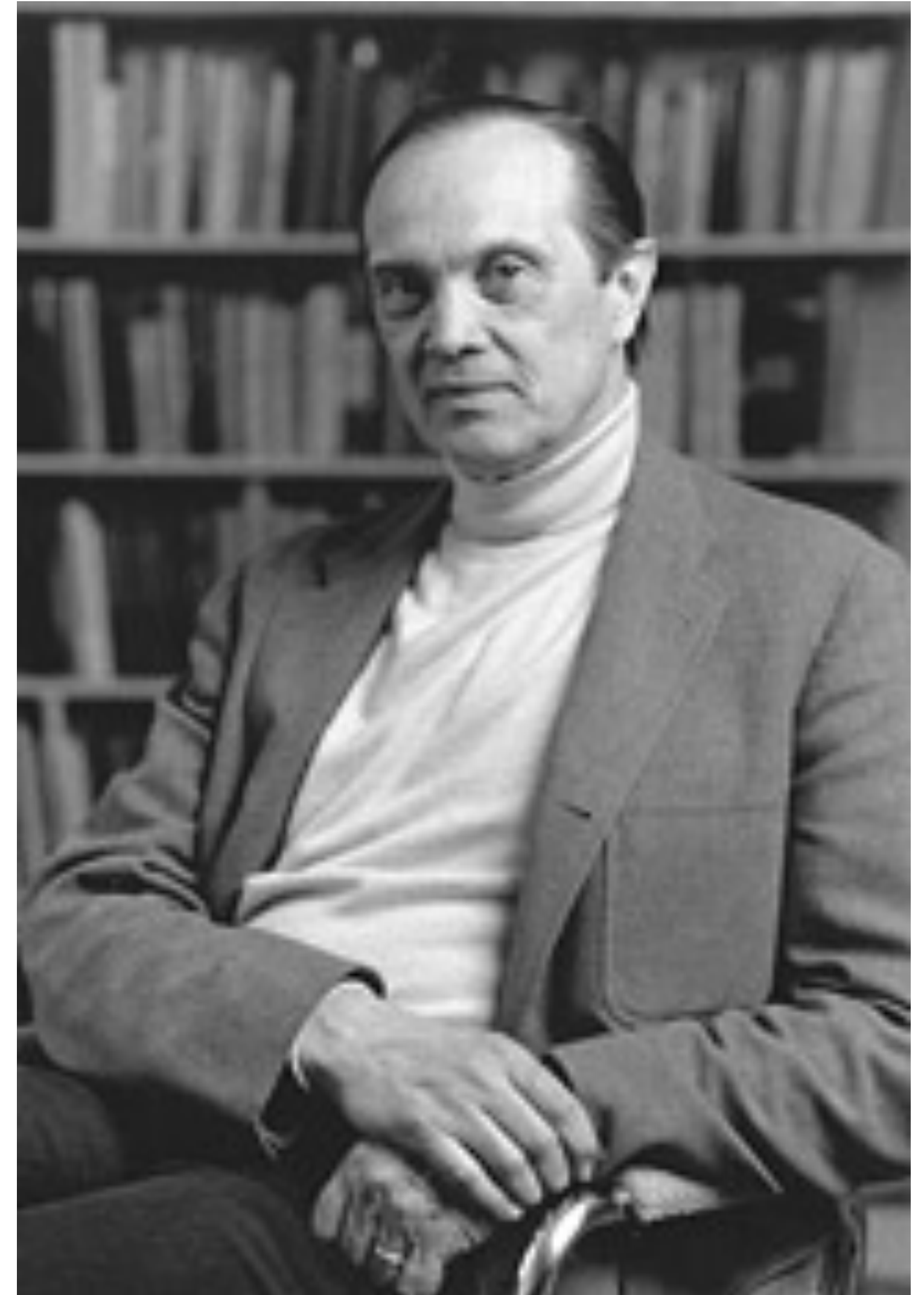
- standard alphabetical procedures organize lexical information and put together words that are spelled alike.
 - scatter words with similar or related meanings haphazardly through the list.
 - there is no obvious alternative, no other simple way for readers to find the word they are looking for.

on-line dictionaries

- computers can search such alphabetical lists much faster than people can: a dictionary entry can be available as soon as the target word is selected or typed into the keyboard.
- however, it quickly becomes apparent that it is grossly inefficient to use these powerful machines as little more than rapid page-turners.

Princeton, 1985

- in 1985 a group of psychologists and linguists at Princeton University undertook to develop a lexical database.
 - the initial idea was to provide an aid to use in **searching dictionaries conceptually**, rather than merely alphabetically.
 - **it was to be used in close conjunction with an on-line dictionary of the conventional type.**
- as the work proceeded, however, it demanded a more ambitious formulation of its own principles and goals.



George Miller - Photo
by John T. Miller

four categories

- the most obvious difference between WordNet and a standard dictionary is that WordNet divides the lexicon into four categories:
 - nouns,
 - verbs,
 - adjectives, and
 - adverbs.

categories organization

- WordNet is an attempt to organize lexical information in terms of *word meanings*.
 - nouns are organized in lexical memory as hierarchies,
 - verbs are organized by a variety of entailment relations,
 - adjectives and adverbs are organized as N -dimensional hyperspaces.
- each of these lexical structures reflects a different way of categorizing experience; attempts to impose a single organizing principle on all syntactic categories would badly misrepresent the psychological complexity of lexical knowledge.

the Lexical Matrix



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lexicalized concept & utterance

- lexical semantics begins with a recognition that a *word* is a conventional association between a *lexicalized concept* and an *utterance* that plays a syntactic role.
- since the word "word" is commonly used to refer both to the *utterance* and to its *associated concept*, discussions of this lexical association are vulnerable to terminological confusion.
 - to reduce ambiguity, "*word form*" will be used here to refer to the *physical utterance* or inscription and "*word meaning*" to refer to the *lexicalized concept* that a form can be used to express.

lexical matrix

Word Meanings	Word Forms				
	F_1	F_2	F_3	$\cdot \cdot \cdot$	F_n
M_1	$E_{1,1}$	$E_{1,2}$			
M_2		$E_{2,2}$			
M_3			$E_{3,3}$		
$\vdots$				$\cdot \cdot \cdot$	
M_m					$E_{m,n}$

- word forms are imagined to be listed as headings for the columns; word meanings as headings for the rows.
- an entry in a cell of the matrix implies that the form in that column can be used (in an appropriate context) to express the meaning in that row.

lexical matrix

Word Meanings	Word Forms				
	F_1	F_2	F_3	$\cdot \cdot \cdot$	F_n
M_1	$E_{1,1}$	$E_{1,2}$			
M_2		$E_{2,2}$			
M_3			$E_{3,3}$		
$\vdots$				$\cdot$	
M_m				$\cdot$	$E_{m,n}$

- entry $E_{1,1}$ means that word form F_1 can be used to express word meaning M_1 .
- if there are **two entries in the same column**, the word form is **polysemous**;
- if there are **two entries in the same row**, the two word forms are **synonyms** (relative to a context).

synonym sets

Word Meanings	Word Forms				
	F_1	F_2	F_3	$\dots$	F_n
M_1	$E_{1,1}$	$E_{1,2}$			
M_2		$E_{2,2}$			
M_3			$E_{3,3}$		
$\vdots$				$\ddots$	
M_m					$E_{m,n}$

- the word meaning M_1 can be represented by simply listing the word forms that can be used to express it: $\{F_1, F_2, \dots\}$.
 - someone who knows that *board* can signify either a piece of lumber or a group of people assembled for some purpose will be able to pick out the intended sense with no more help than *plank* or *committee*.
 - the synonym sets, $\{\textit{board}, \textit{plank}\}$ and $\{\textit{board}, \textit{committee}\}$ can serve as unambiguous designators of these two meanings of board.

synsets

- these synonym sets (*synsets*) do not explain what the concepts are; they merely signify that the concepts exist.
- people who know English are assumed to have *already acquired the concepts*, and are expected to *recognize* them from the words listed in the synset.

gloss

- since English is rich in synonyms, synsets are often sufficient for *differential purposes*.
- sometimes, however, an appropriate synonym is not available, in which case the *polysemy can be resolved by a short gloss*, e.g.,
{board, (a committee having supervisory powers; "the board has seven members")}.
- a gloss can serve to *differentiate this sense of board from the others*; it can be regarded as a synset with a single member.

semantic relations

- WordNet is *organized by semantic relations*. since a semantic relation is a relation between meanings, and since meanings can be represented by synsets, it is natural to think of *semantic relations as pointers between synsets*.
- it is characteristic of *semantic relations* that they *are symmetric*: if there is a semantic relation R between meaning $\{x, x', \dots\}$ and meaning $\{y, y', \dots\}$, then there is also a relation R' between $\{y, y', \dots\}$ and $\{x, x', \dots\}$.

semantic relations



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synonymy

- the **most important** relation for WordNet is *similarity of meaning*, since the ability to judge that relation between word forms is a prerequisite for the representation of meanings in a lexical matrix.
- two expressions are *synonymous* if the *substitution* of one for the other *never changes the truth value* of a sentence in which the substitution is made

synonymy

- by that definition, true synonyms are rare, if they exist at all.
 - a **weakened** version of this **definition** would make synonymy relative to a context: two expressions are synonymous **in a linguistic context C** if **the substitution of one for the other in C does not alter the truth value**.
 - e.g., the substitution of *plank* for *board* will seldom alter truth values in carpentry contexts.

antonymy

the word rich is an antonym of the word poor

- antonymy turns out to be surprisingly difficult to define. the antonym of a word x is sometimes *not- x* , but not always.
 - *rich* and *poor* are antonyms, but to say that someone is not rich does not imply that they must be poor; many people consider themselves neither rich nor poor.
- antonymy is actually quite complex, yet speakers of English have little difficulty recognizing antonyms when they see them.

antonymy is a *lexical* relation

- antonymy is a *lexical* relation between word forms, not a *semantic* one between word meanings.
 - e.g., the meanings {*rise*, *ascend*} and {*fall*, *descend*} may be *conceptual opposites*, but they are not antonyms; [*rise/fall*] are antonyms and so are [*ascend/descend*], but most people hesitate and look thoughtful when asked if *rise* and *descend*, or *ascend* and *fall*, are antonyms.
- such facts make apparent the need to distinguish between *semantic relations between word forms* and *semantic relations between word meanings*.



hyponymy

- unlike synonymy and antonymy, which are lexical relations between word forms, *hyponymy/hypernymy* is a semantic relation between word meanings: e.g., {*maple*} is a *hyponym* of {*tree*}, and {*tree*} is a *hyponym* of {*plant*}.
- a concept represented by the synset {*x*, *x'*, . . . } is said to be a *hyponym* of the concept represented by the synset {*y*, *y'*, . . . } if native speakers of English accept sentences constructed from such frames as "*An x is a (kind of) y*".

hyponymy

- hyponymy is *transitive* and *asymmetrical*, and, since there is normally a single superordinate, it generates a hierarchical semantic structure, in which a *hyponym* is said to be *below its superordinate*.
- a hyponym *inherits all the features* of the more generic concept *and adds at least one feature* that distinguishes it from its superordinate and from any other hyponyms of that superordinate.

meronymy

- another relation is the *part-whole* (or HASA) relation.
- a concept represented by the synset $\{x, x', \dots\}$ is a *meronym* of a concept represented by the synset $\{y, y', \dots\}$ if native speakers of English accept sentences constructed from such frames as "A y has an x (as a part)" or "An x is a part of y ".
- the meronymic relation is *transitive* and *asymmetrical*, and can be used to construct a part hierarchy (with some reservations, since a meronym can have many holonyms).

a complex network of relations

- these and other similar relations can be represented in WordNet by **pointers** (labeled arcs) from one synset to another.
- these **relations** represent associations that **form a complex network**;
 - knowing **where a word is situated** in that network is an important part of knowing the word's meaning.

Nouns in WordNet: A Lexical Inheritance System



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inheritance (distinguishing features)

- *definitions* of common nouns typically give a superordinate term plus distinguishing features; that information provides the basis for organizing noun files in WordNet.
 - each word *inherits* the distinguishing features of all its superordinates.
- three types of distinguishing features are discussed: *attributes* (*modification*), *parts* (*meronymy*), and *functions* (*predication*).
- coverage is partitioned into twenty-five topical files, each of which deals with a different primitive semantic component.

superordinate + distinguishers

- take one meaning of the noun *tree*, the sense having to do with *trees* as *plants*. conventional dictionaries define this sense of *tree* by some such gloss as: *a plant that is large, woody, perennial, and has a distinct trunk.*

albero¹ [àl-be-ro] s.m.

1 Ogni pianta perenne con fusto legnoso (tronco) ramificato: *a. da frutto*; *piantare un a.*; *chioma dell'a.*

2 È usato nella denominazione di molte varietà di piante || *a. del latte*, albero presente nella fascia tropicale americana, dalla cui corteccia incisa fuoriesce un lattice commestibile | *a. del pane*, albero tropicale con frutti ricchi di amido | *a. di Giuda*, pianta con fiori rossi a grappolo

3 *fig.* Numerosi anche gli impieghi simbolici || *a. della cuccagna*, palo insaponato alla cui cima sono appesi vari doni, perlopiù alimentari, destinati a chi riesce a raggiungerli arrampicandosi | *a. di Natale*, pianta di abete o di altro tipo, anche finta, addobbata nel periodo natalizio | *a. genealogico*, rappresentazione grafica dei rapporti di discendenza a forma di albero stilizzato | *a. sintattico*, *in linguistica*, rappresentazione grafica dei rapporti tra i costituenti di una frase

4 *mar.* Antenna, palo verticale in legno, metallo o fibra sintetica fissata allo scafo, che sorregge pennoni, vele ecc.: *a. maestro* o *di maestra*

5 *mecc.* Organo di varie macchine motrici atto a trasmettere il moto e la potenza, costituito da un asse rotante a sezione circolare **SIN** *asse*: *a. motore* || *a. a camme*, nei motori a scoppio, diesel ecc., con camme d'azionamento delle valvole || *a. di trasmissione*, negli autoveicoli, quello che trasmette il moto dal cambio al differenziale

tree |trē|

noun

1 a woody perennial plant, typically having a single stem or trunk growing to a considerable height and bearing lateral branches at some distance from the ground. Compare with **SHRUB** ¹.

- (in general use) any bush, shrub, or herbaceous plant with a tall erect stem, e.g., a banana plant.

2 a wooden structure or part of a structure.

- archaic or poetic/literary the cross on which Jesus Christ was crucified.
- archaic a gallows or gibbet.

3 a thing that has a branching structure resembling that of a tree.

- (also **tree diagram**) a diagram with a structure of branching connecting lines, representing different processes and relationships.

verb (**trees** , **treed** , **tree•ing**) [trans.]

force (a hunted animal) to take refuge in a tree.

- informal force (someone) into a difficult situation.



leaves of familiar trees of
North America

cosa manca?



no full specification

- this definition does not say, for example, that *trees have roots*, or that *they consist of cells having cellulose walls*, or even that *they are living organisms*.
- of course, if you *look up the superordinate term, plant*, you *may find that kind of information*, unless you make a mistake and choose the definition of *plant* that says it is a *place* where some product is manufactured.
- there is *nothing in the definition of tree* that *specifies which sense of plant is the appropriate superordinate*.

no info on coordinate terms

- second, this definition of *tree* contains **no information about coordinate terms**.
- the existence of other kinds of *plants* is a plausible conjecture, but no help is given in finding them.
 - a reader curious about coordinate terms has little alternative but to **scan the dictionary from A to Z**, noting along the way each occurrence of a definition with the superordinate term plant.

no downward pointers to hyponyms

- a similar challenge faces a reader who is interested in knowing the *different kinds of trees*.
 - in addition to looking through the dictionary for familiar trees as *pine* or *maple* or *oak*, a reader might wish to know which trees are *deciduous*, which are *hardwoods*, or how many different kinds of *conifers* there are.
- prototypical definition points upward, to a superordinate term, not sideways to coordinate terms or downward to hyponyms.

a web of words

- every dictionary probably contains a few vacuous circles, instances where word W_a is used to define word W_b and W_b is also used to define W_a .
- the fundamental design that lexicographers try to impose on the semantic memory for nouns is *not a circle, but a tree* (in the sense of tree as a graphical representation).
 - it is a defining property of tree graphs that they branch from a single stem without forming circular loops.



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psycholinguistic assumptions



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psycholinguistic evidence

- since WordNet is supposed to be organized according to principles governing human lexical memory, the **decision to organize the nouns as an inheritance system reflects a psycholinguistic judgment about the mental lexicon.**
- psycholinguistic evidence that knowledge of nouns is organized hierarchically comes from the **ease with which people handle anaphoric nouns and comparative constructions.**

superordinate-hyponym

- **superordinate** nouns can serve **as anaphors** referring back to **their hyponyms**:
 - in such constructions as *He owned a rifle, but the gun had not been fired*, it is immediately understood that the **gun** is an **anaphoric noun** with a **rifle** as its antecedent.
- **superordinates and their hyponyms cannot be compared**:
 - both *A rifle is safer than a gun* and *A gun is safer than a rifle* are immediately recognized as semantically anomalous.
- such judgments demand an **explanation in terms of hierarchical semantic relations**.

experimental evidence

- some authors assumed that *reaction times* can be used to indicate the number of hierarchical levels separating two meanings.
- they observed that it takes less time to respond *true* to "*A canary can sing*" than to "*A canary can fly*" and still more time is required to respond *true* to "*A canary has skin*".
 - in this example, it is assumed that *can sing* is stored as a feature of *canary*, *can fly* as a feature of *bird*, and *has skin* as a feature of *animal*.

Collins and Quillian, Retrieval time from semantic memory, 1969.

experimental evidence

- if all three features had been stored directly as features of canary, they could all have been retrieved with equal speed.
- Hypothesis: the reaction times are not equal because additional time is required to retrieve *can fly* and *has skin* from the superordinate concepts.
 - from such observations it was concluded that generic information is not stored redundantly, but is retrieved when needed.

open issues

- there is general consensus that English common nouns are organized hierarchically in semantic memory, but *whether generic information is inherited or is stored redundantly is still moot.*
- the publication of Collins and Quillian's (1969) experiments stimulated considerable research, in the course of which *a number of problems were raised.*
 - for example, according to Quillian's theory, *robin* and *ostrich* share the same kind of semantic link to the superordinate *bird*, yet "*A robin is a bird*" is confirmed more rapidly than is "*An ostrich is a bird*".

open issues

- from these and similar results, many scholars concluded that Quillian was wrong, that **semantic memory for nouns is not organized as an inheritance system**.
- an alternative conclusion -the conclusion on which WordNet is based- is that **the inheritance assumption is correct, but that reaction times** do not measure what Collins and Quillian, and other experimentalists assumed they did.
 - perhaps reaction times **indicate a pragmatic rather than a semantic distance**: a **difference in word use**, rather than a difference in word meaning.



semantic components



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a multi-hierarchy

- the nouns are partitioned into a set of semantic primes, that select a (relatively small) number of generic concepts and each one acts as the unique beginner of a separate hierarchy.
- these multiple hierarchies correspond to relatively distinct semantic fields, each with its own vocabulary.
- that is to say, since the features that characterize a unique beginner are inherited by all of its hyponyms, a unique beginner can be regarded as a *primitive semantic component* of all words in its hierarchically structured semantic field.

25 unique beginners (nouns)

{*act, action, activity*}

{*animal, fauna*}

{*artifact*}

{*attribute, property*}

{*body, corpus*}

{*cognition, knowledge*}

{*communication*}

{*event, happening*}

{*feeling, emotion*}

{*food*}

{*group, collection*}

{*location, place*}

{*motive*}

{*natural object*}

{*natural phenomenon*}

{*person, human being*}

{*plant, flora*}

{*possession*}

{*process*}

{*quantity, amount*}

{*relation*}

{*shape*}

{*state, condition*}

{*substance*}

{*time*}

hierarchies

- to decide what these primitive semantic components should be, one important criterion is that, **collectively, they should provide a place for every English noun.**
- **these hierarchies** vary widely in size and are not mutually exclusive —some cross-referencing is required— but **on the whole** they **cover distinct conceptual and lexical domains.**
- **seven of the components**, for example, were concerned with *living or non-living things*; they could be arranged hierarchically and **a small 'Tops' file was created in order to include these semantic relations in the system.**

hyponymic relations among seven unique beginners
denoting different kinds of tangible things.

{thing, entity}

{living thing, organism}

{non-living thing, object}

{plant, flora}

{animal, fauna}

{person, human being}

{natural object}

{artifact}

{substance}

{food}

distinguishing features



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generic concepts

- these hierarchies of nominal concepts are said to *have a level, somewhere in the middle, where most of the distinguishing features are attached*. it is referred to as the basic level, and the nominal concepts at this level are called *basic-level categories* or *generic concepts*.
- for concepts at the basic level, *people can list many distinguishing features* (Rosch, 1975).
 - *above* the basic level, descriptions are *brief* and *general*.
 - below the base level, little is added to the features that distinguish basic concepts.

attributes

- although the *overall structure* of noun hierarchies is generated by the *hyponymy* relation, *details* are given by the *features* that distinguish one concept from another.
 - e.g., a *canary* is a *bird* that is *small*, *colorful*, *sings*, and *flies*,
- so *not only must canary be entered as a hyponym of bird*, from which it inherits the fact that it *has a beak and wings with feathers*.
- but the *attributes of small size and bright color must also be included*, as well as the *activities of singing and flying*.

attributes

- in order to make all of this information available when canary is activated, *it must be possible to associate canary appropriately with at least three different kinds of distinguishing features.*
 1. *Attributes: small, yellow*
 2. *Parts: beak, wings*
 3. *Functions: sing, fly*
- each type of distinguishing feature must be *treated differently.*
 - *attributes* are given by *adjectives*, *parts* by *nouns*, and *functions* by *verbs*.

attributes and modification



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attributes as modifiers

- *values of attributes are expressed by adjectives.*
 - e.g., *size* and *color* are attributes of canaries: *the size of canaries can be expressed by the adjective small*, and the usual *color* of canaries can be expressed by the adjective *yellow*.
- *adjectives* are said to *modify* *nouns*, or *nouns* are said to *serve as arguments* for attributes:
Size(canary) = small.

attributes as *possible* modifiers

- the *attributes* associated with a noun *are reflected in the adjectives that can normally modify that name.*
 - e.g., *a canary can be hungry or satiated* because hunger is a feature of animals and canaries are animals, but
 - *a stingy canary or a generous canary could only be interpreted metaphorically*, since generosity is not a feature of animals in general, or of canaries in particular.

parts and meronymy



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meronymy

- the *part-whole relation* between nouns is generally considered to be a semantic relation, called meronymy, comparable to *synonymy*, *antonymy*, and *hyponymy*.
- the meronymy relation has an *inverse*: if W_m is a meronym of W_h , then W_h is said to be a *holonym* of W_m .

meronymy

- meronyms *are distinguishing features* that hyponyms *can inherit*. *meronymy* and *hyponymy* become intertwined in complex ways.
 - e.g., if *beak* and *wing* are meronyms of *bird*, and if *canary* is a hyponym of *bird*, then, by inheritance, *beak* and *wing* must also be meronyms of *canary*.
 - although the connections may appear complex when dissected in this manner, they are *rapidly deployed in language comprehension*.

asymmetric and transitive relation...

- the "part of" relation is *asymmetric* and (with reservations) *transitive*.
- in many instances *transitivity seems to be limited*.
 - e.g., *handle* is a meronym of *door* and *door* is a meronym of *house*, yet it sounds odd to say "*The house has a handle*" or "*The handle is a part of the house*".
 - e.g., "*The branch is a part of the tree*" and "*The tree is a part of a forest*" do not imply that "*The branch is a part of the forest*" because the *branch/tree* relation is not the same as the *tree/forest* relation.

"part of" relations

- in literature **six types of meronyms** have been individuated:
 - *component-object* (branch/tree),
 - *member-collection* (tree/forest),
 - *portion-mass* (slice/cake),
 - *stuff-object* (aluminum/airplane),
 - *feature-activity* (paying/shopping), and
 - *place-area* (Princeton/New Jersey).
- Chaffin, Hermann, and Winston (1988) add a **seventh**:
 - *phase-process* (adolescence/growing up).



functions and predication



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functions

- a *functional feature* of a nominal concept is a description of something that *instances of the concept normally do*, or that *is normally done with or to them*.
- this usage feels more natural in some cases than in others.
 - e.g., it seems natural to say that *the function of a pencil is to write* or *the function of a knife is to cut*, but to say that *the function of a canary is to fly or to sing* seems a bit forced.

functions

- all the features of nominal concepts that are described by verbs or verb phrases.
 - nominal concepts can play various *semantic roles* as arguments of the verbs that they co-occur with in a sentence: *instruments* (*knife-cut*), *materials* (*wool-knit*), *products* (*hole-dig; picture-paint*), *containers* (*box-hold*), etc.

function as feature of meaning

- there are also linguistic reasons to assume that a *thing's function is a feature of its meaning*.
- consider the problem of defining the adjective *good*.
 - a *good pencil* is one that writes easily, a *good knife* is one that cuts well, a *good light* is one that illuminates brightly, and so on.
 - as the head noun changes, *good* takes on a sequence of meanings: writes *easily*, cuts *well*, etc.
- it is unthinkable that all of these different meanings should be listed in a dictionary entry. how should this problem be handled?

function as feature of meaning

- a surprising consequence of this formulation is that **an object that is not an *X*** can be said to be a good *X* if it performs well the function that *X*'s normally perform.
 - e.g., calling a *box* a *chair* does not make it one, yet a person who sits on a box can say "*This box is a good chair*" and so indicate that *the box is performing well the function that chairs are expected to perform*.

semantic roles

- *the semantic role of nouns* like *hammer*, *wallpaper*, or *box* tend to be the same wherever they occur in sentences, *independent of their grammatical role*.
 - in both *John hit the mugger with a hammer* and *The hammer hit him on the head*, the *semantic role of hammer* is that of an *instrument*.
- this consistency in mapping onto the same semantic role independently of syntax is *not a feature of all nominal concepts*, however: *what is the function of apple or cat?*

english verbs as a semantic net



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verbs

- verbs are arguably the **most important lexical and syntactic category** of a language.
 - in most languages **all sentences must contain at least one verb**, but, as grammatical sentences with "dummy" subjects like *It is snowing* show, they need not contain a (referential) noun.
 - the verb provides the **relational and semantic framework for its sentence**. its predicate-argument structure (or *subcategorization frame*) specifies the **possible syntactic structures** of the sentences in which it can occur.

subcategorization frames

- the linking of **noun arguments with thematic roles** or cases, such as *INSTRUMENT*, determines the **different meanings of the events or states denoted by the sentence**
- the **selectional restrictions** specify the semantic properties of the noun classes that can flesh out the frame.
- this **syntactic and semantic information** is generally thought to be **part of the verb's lexical entry**, that is to say, part of the information about the verb that is stored in a speaker's mental lexicon.

far fewer verbs than nouns...

- even though grammatical English sentences require a verb though not necessarily a noun, *the language has far fewer verbs than nouns*.
 - e.g., the Collins English Dictionary lists 43,636 different nouns and 14,190 different verbs.
- *verbs are more polysemous than nouns*: the nouns in Collins have on the average 1.74 senses, whereas verbs average 2.11 senses.

high mutability of verbs

- the higher polysemy of verbs suggests that *verb meanings* are *more flexible* than noun meanings.
 - verbs can change their *meanings depending* on the kinds of *noun arguments with which they co-occur*;
 - viceversa, the meanings of nouns tend to be more stable in the presence of different verbs.

polysemy due to abstract/concrete obj

- the **most frequently used verbs** (*have, be, run, make, set, go, take, and others*) **are also the most polysemous**, and their meanings often depend heavily on the nouns with which they co-occur.
 - e.g., dictionaries differentiate between the senses of have in sentences like *I have a Mercedes* and *I have a headache*.
 - the difference is less due to the polysemy of *have*, however, than to the **concrete** or **abstract nature of its objects**.

strike: colpire
throb: battere (di pulsazione)
baffle: sconcertare
stir: mescolare
whisk: frullare, tritare

beat

- meaning differences can be determined by different **elaborations of one or two common core components shared by most senses** of the verb (e.g., *beat*).
 - **different senses of beat** occur in very different semantic domains: {*beat, strike, hit*} is a **contact** verb; {*beat, throb, pulse*} is a **motion** verb; {*beat, defeat*} is a **competition** verb; {*beat, flog, punish*} and {*beat, circumvent (the system)*} are verbs in the domain of **social interaction**; {*beat, shape, do metalwork*} is a **creation** verb; {*beat, baffle*} is a **cognition** verb; {*beat, stir, whisk*} belongs to the domain of **cooking** verbs; and {*beat, mark*} is a kind of motion performed to indicate the **counts in music**.
- although most of these verbs seem to share a semantic component of *CONTACT* or *IMPACT*, the differences illustrate how flexible these core meanings can be.

events/actions verbs

- as of April 2023, the database contains about 150,000 words organized in over 117,000 synsets for a total of 207,000 word-sense pairs.
- verbs are divided into 15 files, largely on the basis of semantic criteria. all but one of these files correspond to semantic domains: verbs of *bodily care and functions, change, cognition, communication, competition, consumption, contact, creation, emotion, motion, perception, possession, social interaction*, and *weather* verbs.

states verbs

- another file contains verbs referring to **states**, such as *suffice*, *belong*, and *resemble*, that could not be integrated into the other files.
- the **verbs** in this **latter group** do not constitute a **semantic domain**, and **share no semantic properties** other than that they refer to states.

hyponymy

- the **semantic distinction** between two verbs is **different from** the features that distinguish two nouns in a **hyponymic relation**.
 - *an x is a y*, is not suitable for verbs, because it requires that x and y be nouns: *to amble is a kind of to walk* is not a felicitous sentence.

semantic elaborations: conflations

- an examination of 'verb hyponyms' and their superordinates shows that *lexicalization involves many kinds of semantic elaborations across different semantic fields.*
 - e.g., *motion verbs* could be thought of as *conflations of move and such semantic components as MANNER and CAUSE,* exemplified by *slide* and *pull*, respectively.
- some verbs refer to different degrees of *INTENSITY* of the action or state (*drowse, doze, sleep*).

troponymy

- the many different kinds of elaborations that distinguish a 'verb hyponym' from its superordinate have been merged into troponymy (from the Greek *tropos*: manner or fashion).
- the troponymy relation between two verbs can be expressed by the formula *To V_1 is to V_2 in some particular manner*.
- troponyms can be related to their superordinates along many semantic dimensions and subsets of particular kinds of manners tend to cluster within a given semantic field.

hierarchy

- in trying to construct verb taxonomies using the troponymy relation, it became apparent that **verbs cannot easily be arranged into the kind of tree structures onto which nouns are mapped.**
- within a single semantic field it is frequently the case that **not all verbs can be grouped under a single unique beginner.**

hierarchy

- some semantic fields must be represented by several independent trees.
 - *motion verbs*, for example, have *two top nodes*, {*move, make a movement*}, and {*move, travel*}.
 - *verbs of possession* can be traced up to the three verbs {*give, transfer*}, {*take, receive*}, and {*have, hold*}; for the most part, *their troponyms encode ways in which society has ritualized the transfer* of possessions: *bequeath, donate, inherit, usurp, own, etc.*

hierarchy

- verb hierarchies tend to have a **more shallow, bushy structure than nouns**; in few cases does the number of hierarchical levels exceed four.
 - virtually every verb taxonomy shows what might be called a *bulge*, that is to say, **a level far more richly lexicalized than the other levels** in the same hierarchy.

